A Review on Hand Gesture Recognition: Datasets, Methods, and Applications

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***Abstract*—Hand Gesture Recognition (HGR) has emerged as a key area in computer vision and human-computer interaction, enabling natural and intuitive communication between humans and machines. This paper presents a comprehensive review of existing HGR approaches, datasets, applications, and current research challenges. We categorize methods into vision-based and sensor-based approaches, outline major datasets, and discuss future research directions.**

***Index Terms*—Hand Gesture Recognition, Deep Learning, Computer Vision, Human-Computer Interaction, Review.**

I. Introduction

Hand gesture recognition enables machines to interpret human hand movements as commands or information. Applications range from sign language recognition to virtual reality and robot control. This review aims to summarize the current state of research, categorize methods, present benchmark datasets, and highlight challenges.

II. Taxonomy of Hand Gesture Recognition

Hand gesture recognition can be categorized based on:

- **Gesture type:** Static (single pose) vs. Dynamic (movement over time)
- **Acquisition method:** Vision-based (camera input) vs. Sensor-based (wearable devices)
- **Application domain:** Sign language interpretation, VR/AR, robotics, healthcare

III. Datasets for HGR

Table I lists commonly used datasets in HGR research.

TABLE I
Common Hand Gesture Recognition Datasets

Dataset	Type	#Classes	Acquisition	Year
LeapGestRecog	Static	10	Webcam	2017
SHREC 2017	Dynamic	14	Depth Camera	2017
NVIDIA HandNet	Static	10	RGB Camera	2020

IV. Methods and Techniques

A. Traditional Machine Learning

Older approaches relied on handcrafted features such as Histogram of Oriented Gradients (HOG), Scale-Invariant Feature Transform (SIFT), and skin color segmentation, combined with classifiers like SVM, KNN, or Hidden Markov Models.

B. Deep Learning Approaches

Convolutional Neural Networks (CNNs) are widely used for static gestures, while 3D CNNs, RNNs, and Transformer architectures are applied for dynamic gesture recognition.

C. Sensor-Based Approaches

These use data from accelerometers, gyroscopes, or electromyography (EMG) sensors embedded in wearable devices.

V. Applications

- **Sign Language Recognition**
- **Gaming and VR/AR Interfaces**
- **Robot Control**
- **Healthcare and Rehabilitation**

VI. Challenges and Limitations

- Sensitivity to lighting and background
- Occlusion and overlapping objects
- User variability (skin tone, hand size)
- Real-time processing constraints

VII. Future Research Directions

Future work may focus on:

- Multimodal fusion of vision and sensor data
- Lightweight architectures for mobile devices
- Large-scale, diverse datasets
- Few-shot and zero-shot gesture recognition

VIII. Conclusion

This review summarizes the key developments in hand gesture recognition, outlines the strengths and weaknesses of existing methods, and proposes directions for future research.

References

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